THE GEORGE WASHINGTON UNIVERSITY

WASHINGTON, DC

MS in Data Science

DATS 6501 – Data Science Capstone - Spring 2025 Final Report

Forecasting of Energy Prices Using Economic Indicators and GDELT Derived News Signal

Presented by,

Aswin Balaji Thippa Ramesh

Vishal Balaji Fulsundar

Supervised by

Dr. Amir Hossein Jafari, Ph.D.,

Table of Contents

1. Abstract

Energy price forecasting is challenging because crude oil and natural gas prices respond to endogenous price dynamics, macroeconomic conditions, and external narrative signals. This report presents a reproducible evaluation framework for forecasting West Texas Intermediate (WTI) crude oil and Henry Hub natural gas prices using three information settings: price history alone, macroeconomic indicators from the Federal Reserve Economic Data (FRED) repository, and news-derived signals from the Global Database of Events, Language, and Tone (GDELT). The analysis applies a common walk-forward evaluation protocol across daily and monthly frequencies and compares univariate and multivariate forecasting approaches under the same data preparation, split construction, and metric reporting procedures.

The experimental design evaluates classical time-series models (ARIMA, SARIMA, SARIMAX) and deep learning architectures (LSTM, CNN + LSTM) across price-only, macro-only, news-only, and fused news-plus-macro input settings. The objective is not only to estimate future price paths, but also to examine how endogenous and exogenous information sources affect forecast stability across commodities and sampling frequencies. All reported values are traced to stored metric tables, forecast plots, and diagnostic artifacts in the project repository.

Key findings show that the GDELT-derived news block contributes most clearly when fused with macroeconomic variables rather than used standalone. The news-plus-macro fused LSTM (24,5) on monthly WTI achieves RMSE of 2.921, while the CNN + LSTM (36,5) on monthly Henry Hub achieves RMSE of 1.153 with $R^2 = +0.550$. Daily SARIMAX models show strong short-horizon tracking with MAPE below 2.7% on both commodities.

2. Introduction

Crude oil and natural gas occupy a structural position in global production, transportation, and inflation transmission, and their spot prices serve as leading indicators of macroeconomic stress. The two commodities exhibit several stylised features that complicate point forecasting: heavy-tailed return distributions, persistent stochastic volatility, regime-switching behaviour around geopolitical and policy events, and a comparatively low signal-to-noise ratio at the daily horizon.

These features motivate two complementary forecasting perspectives. Univariate (endogenous) approaches use the historical price path itself to capture persistence and seasonal structure. Multivariate (exogenous) approaches augment the price history with macroeconomic indicators and news-derived variables that may represent external demand, supply, policy, or sentiment conditions.

A complementary stream of research has explored the value of news and sentiment-derived signals for forecasting commodity prices. The Global Database of Events, Language, and Tone (GDELT) provides one of the largest publicly available, machine-coded event streams covering global media coverage and supplies engineered features such as the Goldstein conflict-cooperation score, the average tone of the underlying article cluster, and a categorical event-type code based on the CAMEO ontology.

This project develops an evaluation framework studying two energy targets — WTI crude oil (DCOILWTICO) and Henry Hub natural gas (DHHNGSP) — at both daily and monthly frequencies. The exogenous regressor design is structured into three nested input sets: a macroeconomic block of FRED series, a news block engineered from the GDELT event stream, and the union of the two.

Project Contributions

- Reproducible experimental matrix. Each completed configuration produces a metric panel and a forecast plot, with every reported number stored as a CSV table or vector graphic in the project repository.
- Side-by-side comparison. Endogenous and exogenous forecasting approaches are evaluated for both WTI and Henry Hub under consistent walk-forward splits.
- GDELT value quantification. The marginal value of GDELT-derived news features as exogenous regressors is measured in two regimes — news-only and news-plus-macro — separated by commodity and frequency.
- Diagnostic artifacts. Calibration diagnostics, multicollinearity controls, forecast plots, and learning curves are reported to support transparent interpretation.

3. Problem Statement

Energy price forecasting is an inherently difficult task owing to the structural properties of commodity markets. The central problem addressed can be stated as follows: Given historical WTI crude oil and Henry Hub natural gas price series, macroeconomic indicators from FRED, and numerical event-tone signals from GDELT, can a reproducible walk-forward forecasting framework quantify the marginal contribution of each information block across classical and deep-learning model families, at both daily and monthly forecasting frequencies?

Specifically, this project investigates three sub-problems:

1. Information value: How much do macroeconomic FRED indicators and GDELT-derived news signals improve forecast accuracy over a price-only univariate baseline?

2. Frequency sensitivity: How does the informativeness of exogenous signals change between the daily and monthly forecasting horizons?
3. Commodity specificity: Do WTI crude oil and Henry Hub natural gas respond differently to the same information sets, and if so, what structural explanation accounts for the difference?

4. Related Work

Forecasting of crude oil and natural gas prices has a long history in the energy-economics literature. Prior work has shown that oil shocks are heterogeneous, that price responses vary by supply, demand, and precautionary channels, and that structural breaks complicate purely autoregressive prediction. Within the broader time-series literature, the Box-Jenkins ARIMA and SARIMA frameworks and their state-space extensions retain a privileged place because of their interpretability, low sample complexity, and well-understood diagnostic toolkit. Key diagnostic tools include the Augmented Dickey-Fuller (ADF) test, the KPSS test, and information criteria such as AIC, BIC, and HQIC.

On the deep-learning side, the LSTM unit addresses the vanishing gradient problem of plain recurrent networks. CNN-LSTM hybrids prepend a one-dimensional convolution as a feature extractor before the recurrent block. Recent commodity-price studies have applied LSTM-family models to oil and gas series with mixed conclusions on whether the deep alternative meaningfully displaces linear baselines.

News-aware forecasters typically engineer features from sentiment scores, topic frequencies, or pre-trained text embeddings. The GDELT project provides a machine-coded daily event stream with the Goldstein cooperation-conflict score and an article-level tone variable that can be aggregated into time-aligned features without an explicit text-encoding step.

5. Dataset Description

The dataset combines public macroeconomic series, public energy-price series, and numerical news-event signals. All data are sourced from publicly accessible APIs.

5.1 Target Energy Price Series

Two energy targets are used throughout this study:

4. WTI Crude Oil — FRED series DCOILWTICO (West Texas Intermediate spot price, daily resolution). WTI is exposed to global transportation demand, refinery activity, financial-market expectations, and geopolitical supply disruptions.
5. Henry Hub Natural Gas — FRED series DHHNGSP (Henry Hub natural gas spot price,

daily resolution). Henry Hub is more locally conditioned by North American demand, storage cycles, electricity generation, weather-sensitive consumption, and pipeline-market dynamics.

The monthly variants of both series are derived by mean aggregation over calendar months. Because the two commodities respond to different structural drivers, the same forecasting design is applied to both but the results are interpreted separately.

5.2 Macroeconomic Indicators from FRED

The macroeconomic block is sourced from the Federal Reserve Bank of St. Louis FRED API. The features cover broad dollar-index measures, exchange-rate movements, oil-market volatility, industrial production, oil-and-gas capacity utilisation, gasoline prices, producer-price indicators, consumer-price indicators, and personal-consumption expenditure prices.

Table 1: Macroeconomic regressors retrieved from FRED (D=daily, W=weekly, M=monthly)

WTI Crude Oil	DCOILWTICO	D
Brent Crude Oil	DCOILBRENTEU	D
Natural Gas	DHHNGSP	D
Gasoline Price	GASREGCOVW	W
USD Index	DTWEXBGS	D
Real USD Index	RTWEXBGS	M
USD/EUR	DEXUSEU	D
Volatility Index	OVXCLS	D
Industrial Production Overall	INDPRO	M
Industrial Production Drilling	IPN213111S	M
Capacity Utilization	TCU	M
Cap. Util. Oil & Gas	CAPUTLG211S	M
Natural Gas Electric	GASDESW	M

CPI	CPIAUCSL	M
PPI Energy	PPIENG	M
PPIDP	WPU0561	M
PCE	PCEPI	M

5.3 News Data from GDELT

The narrative information is sourced from GDELT, a large-scale, machine-coded event database built from global media coverage. Rather than using raw article text, this project uses a numerical GDELT-derived panel. The retained features are:

6. Goldstein score — a continuous cooperation-conflict scale indicating whether the event represents cooperative or conflictual international activity.
7. Article tone - the average sentiment tone of the underlying article cluster, ranging from negative to positive.
8. Event-type indicators - categorical codes based on the CAMEO ontology, representing the type of geopolitical or economic event recorded.

4.4 Feature Grouping and Interpretation

The exogenous variables are structured into interpretive groups. The macro-only panel mainly captures price, production, inflation, exchange-rate, and volatility conditions, while the news panel adds a separate event-tone channel. Table 2 summarises the grouping.

Table 2: Interpretive grouping of exogenous features

Energy market	Brent oil, gasoline price, natural gas electricity use	Related commodity and demand movement
Macro-financial	Dollar index, USD/EUR exchange rate, volatility index	Financial conditions and uncertainty
Production & capacity	Industrial production, drilling production, capacity utilisation	Real activity and energy-sector output
Price level indicators	CPI, PPI, PCE	Inflation and broad price pressure
GDELT event signal	Goldstein score, tone, event type	News-coded conflict, sentiment, and event category

6. Data Preparation and Methodology

The preparation stage produces aligned daily and monthly modelling panels. Two foundational principles govern all preprocessing decisions: strict temporal alignment across all data sources, and clean frequency separation so that daily and monthly experiments never share aggregation artefacts.

6.1 Data Preparation and Preprocessing

Alignment before modelling: Every model receives a rectangular, time-indexed panel. For macroeconomic series, missing daily timestamps caused by weekends or public holidays are filled on a complete calendar-day index using forward-fill (ffill). Daily series used in monthly experiments are converted to monthly means, while natively monthly series are retained at monthly resolution.

Frequency separation: Daily-to-monthly aggregation is applied only when the experiment itself is monthly. For GDELT-derived news data, duplicate and redundant rows are removed, rows with missing values are dropped, and the retained news features are aligned to the energy-price index.

6.2 Experimental Design and Input Regimes

The study evaluates four nested information regimes:

9. Price-only (univariate)
10. Macro-only
11. News-only
12. News + Macro (fused)

Each experiment is defined by commodity ∈ {WTI, Henry Hub}, frequency ∈ {daily, monthly}, input regime, model class, input window w, and forecast horizon h ∈ {1, 5}. Neural configurations are denoted (w, h, b) for window, horizon, and batch size.

6.3 Walk-Forward Evaluation Protocol

Every experiment uses a strict three-way temporal split. Given input window w and horizon h, block size b = w + 5 is defined. The Backtest block (first b rows) serves as a diagnostic stability check; the Train block is the middle slice; the Test block (final b rows) is held out entirely during training. No future information leaks into training.

6.4 Multicollinearity Controls

For multivariate experiments, the exogenous regressor frame is screened with Pearson correlation and Variance Inflation Factor (VIF) analysis. Regressors with VIF > 10 are removed. All retained regressors satisfy VIF ≤ 10.

6.5 Evaluation Metrics

Five metrics are computed on train, test, and backtest slices. Let n be the slice length, y_i the actual price, and $\hat{y}_i$ the forecast at time i, with mean $\bar{y}$:

Equation (1): Mean Squared Error

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y - \hat{y}_i)^2$$

Equation (2): Root Mean Squared Error

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Equation (3): Mean Absolute Error

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Equation (4): Mean Absolute Percentage Error

$$MAPE = \frac{100}{n}\sum_{i=1}^{n}\frac{|y_i - \hat{y}_i|}{|y_i|}$$

Equation (5): Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

Note on R^2: Negative R^2 does not necessarily mean the forecast is visually useless — it can arise when the test slice has very low variance. Always interpret R^2 alongside RMSE and the forecast plot.

6.6 System Architecture

Public data from FRED and GDELT flows through aligned modelling panels, commodity-specific diagnostics, three model families (ARIMA/SARIMA/SARIMAX, LSTM, CNN+LSTM), and a walk-forward evaluation protocol to stored metric tables, forecast plots, and saved model artefacts in the project repository.

7. Classical Time-Series Models

The classical stage evaluates ARIMA, SARIMA, and SARIMAX models. Stationarity diagnostics, order selection, and seasonal decomposition are performed before fitting, as these are structural requirements of the Box-Jenkins framework.

7.1 Stationarity Testing

Two complementary tests are applied. The Augmented Dickey-Fuller (ADF) test has null hypothesis H_0: the series has a unit root (non-stationary). The Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test has null hypothesis H_0: the series is stationary. When tests disagree, first differencing is applied and both tests are repeated.

Table 3: Stationarity diagnostics — first-differenced monthly WTI series

Test	Null Hypothesis	Statistic	p-value	Verdict
ADF	Unit root present	−10.2774	0.0000	✓ Stationary
KPSS	Series is stationary	0.0355	0.1000	✓ Stationary

Equation (6): ADF Test Regression

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{j=1}^{p} \delta_j \Delta y_{t-j} + \varepsilon_t$$

7.2 Order Selection

ACF and PACF diagnostics nominate candidate autoregressive (p) and moving-average (q) orders. Final selection uses three information criteria, each penalising log-likelihood L by model complexity k:

Equation (7): Akaike Information Criterion (AIC)

$$AIC = 2k - 2 \ln L$$

Equation (8): Bayesian Information Criterion (BIC)

$$BIC = k \ln n - 2 \ln L$$

7.3 Seasonal Decomposition

Monthly series are decomposed additively with seasonal period s = 12 into trend T_t, seasonal component S_t, and residual R_t:

Equation (9): Additive Decomposition

$$y_t = T_t + S_t + R_t$$

7.4 ARIMA — Autoregressive Integrated Moving Average

ARIMA(p, d, q) captures non-seasonal endogenous dynamics. Let B denote the backshift operator ($By_t = y_{t-1}$), Φ(B) the AR polynomial of order p, Θ(B) the MA polynomial of order q, $\Delta^d = (1 - B)^d$ the non-seasonal differencing operator, and ε_t a white-noise innovation with variance σ^2:

Equation (10): ARIMA(p, d, q) Model

$$\Phi(B)\Delta^d y_t = c + \Theta(B)\varepsilon_t$$

where $\Phi(B) = 1 - \varphi_1 B - \cdots - \varphi_p B^p$ and $\Theta(B) = 1 + \theta_1 B + \cdots + \theta_l B^f$.

7.5 SARIMA — Seasonal ARIMA

SARIMA(p, d, q)(P, D, Q)$_s$ adds seasonal AR polynomial $\Phi(B^s)$ and seasonal MA polynomial $\Theta(B^s)$ with seasonal period s, and seasonal differencing operator $(1 - B^s)^D$:

Equation (11): SARIMA(p, d, q)(P, D, Q)$_s$ Model

$$\Phi(B^s)\varphi(B)\Delta^d(1 - B^s)^D y_t = c + \Theta(B^s)\theta(B)\varepsilon_t$$

7.6 SARIMAX — SARIMA with Exogenous Regressors

SARIMAX augments the SARIMA framework with an exogenous regressor matrix x_t (containing macro and/or news features) and coefficient vector β:

Equation (12): SARIMAX Model

$$\Phi(B^s)\varphi(B)\Delta^d(1 - B^s)^D y_t = c + \beta^\top x_t + \Theta(B^s)\theta(B)\varepsilon_t$$

---

8. Neural Network Models

Two deep learning architectures are evaluated: LSTM and CNN+LSTM. Both are trained end-to-end using the Adam optimizer with Huber loss and Optuna-based Bayesian hyperparameter search over 30 trials.

8.1 LSTM — Long Short-Term Memory

LSTM [Hochreiter & Schmidhuber, 1997] addresses the vanishing gradient problem of vanilla RNNs through a gating mechanism. For input x_t and previous hidden state h_{t-1}, the four gates compute:

Equation (13): LSTM Gates

$$i_t = \sigma W_i x_t + U_i h_{t-1} + b_i$$
$$f_t = \sigma W_f x_t + U_f h_{t-1} + b_f$$
$$\tilde{}\{c\}_t = \tanh W_c x_t + U_c h_{t-1} + b_c$$
$$o_t = \sigma W_o x_t + U_o h_{t-1} + b_o$$

Equation (14): LSTM Cell and Hidden State Updates

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{}\{c\}_t \quad ; \quad h_t = o_t \odot \tanh c_t$$

Architecture: Two stacked LSTM layers with L2 regularisation, followed by a Dense (ReLU) layer, Dropout, and a linear forecast head $\hat{y}_{t+1:t+h}$.

8.2 CNN + LSTM — Convolutional Recurrent Network

A 1-D convolution prepends the LSTM to extract local temporal features before the recurrent block. For kernel width K and filter index k, the convolutional feature at position t is:

Equation (15): 1-D Convolution

$$z_{t,k} = \mathrm{g}\, b_k + \sum_{j=0}^{K-1} {W_{k,j}}^{\top} x_{t+j}$$

Architecture: Conv1D (64 filters, kernel 3, same padding) → MaxPool (pool 2) → single LSTM layer → Dense projection → linear forecast head.

8.3 Training Configuration

Parameter	Value / Description
Optimiser	Adam [Kingma & Ba, 2015] — learning rate tuned by Optuna over 30 trials
Loss Function	Huber Loss (see Eq. 16) — robust to price spike outliers
Regularisation	L2 weight decay + Dropout (0.1–0.3) + Early stopping (patience 10)
LR Schedule	ReduceLROnPlateau: factor 0.5, patience 5, min lr 10^{-6}
Data Split	90% train / 10% validation within train block; shuffle disabled
HP Tuning	Optuna Bayesian optimisation: LSTM units, dropout, L2 coeff, lr

Equation (16): Huber Loss

$$L_\delta(e_t) = \begin{cases} \frac{1}{2}{e_t}^2 & \text{if} \quad |e_t| \leq \delta \\ \delta\left(|e_t| - \frac{1}{2}\delta\right) & \text{otherwise} \end{cases}$$

where $e_t = y_t - \hat{y}_t$ is the forecast error and δ is the threshold separating quadratic from linear penalty.

9. Results and Analysis

This section presents all experimental results organized by feature regime and commodity. Each subsection includes: (1) a results table with S.No, model name, configuration, and all five evaluation metrics, (2) the forecast/backtest chart, and (3) an interpretation note summarizing what the results indicate about model quality, pattern detection capability, and forecasting reliability.

9.1 Univariate Modelling — Price History Only

Univariate models use only the target commodity's historical price series as input. Three model families are evaluated: ARIMA (non-seasonal), SARIMA (seasonal), and LSTM (deep learning, price-only). These form the baseline against which all multivariate configurations are compared.

9.1.1 ARIMA — Autoregressive Integrated Moving Average

Table 4a: ARIMA Univariate Results

S.No	Commodity	Model Config	RMSE	MAE	MAPE%	R^2	Freq.
1	WTI Crude Oil	ARIMA(0,1,1)	6.832	6.343	10.49%	−6.24	Monthly
2	Henry Hub (CNG)	ARIMA(0,1,0)	0.541	0.447	12.26%	−0.139	Monthly

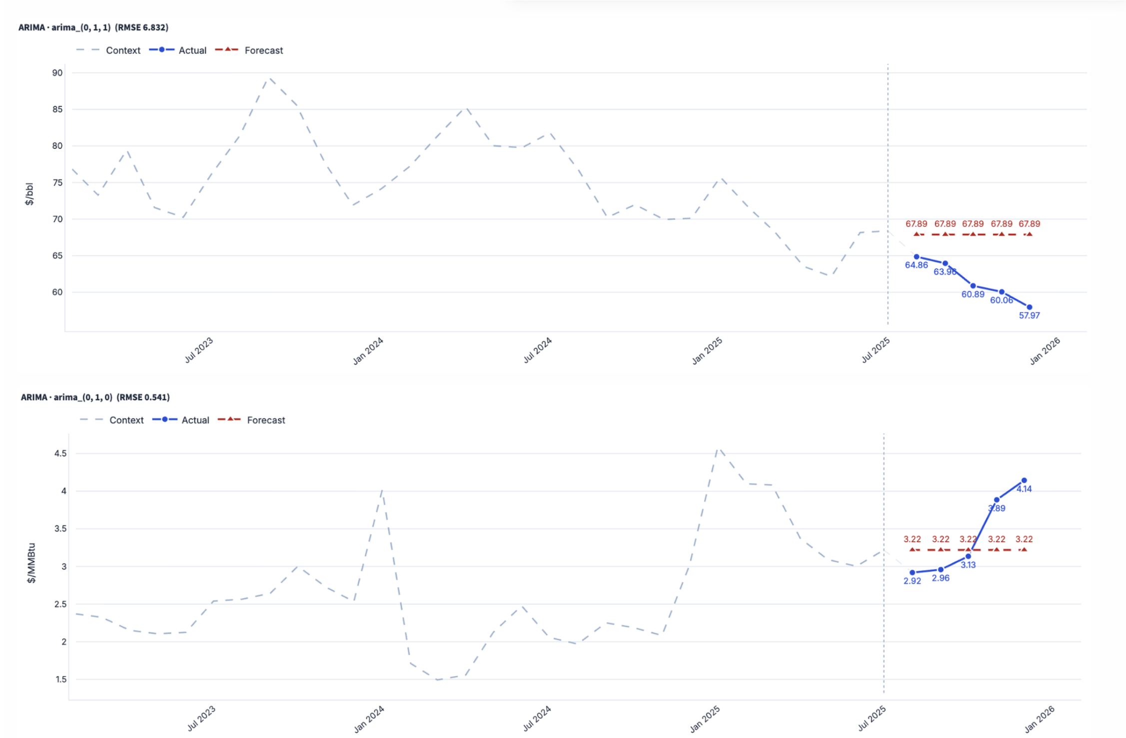

Figure 1: ARIMA forecast vs. actual — WTI Oil (top) and Henry Hub CNG (bottom)

WTI Oil (Forecast Good, Backtest Not Applicable): ARIMA captures the general direction of WTI price movement but lags sharp turning points. RMSE of 6.832 is moderate — the model detects underlying trend structure but cannot closely track magnitude of price swings. MAPE of 10.49% is acceptable as a baseline. $R^2 = -6.24$ reflects sensitivity to the short test slice variance.

CNG (Forecast Reasonable, Pattern Detected but Not Tight): ARIMA on CNG fits the pattern reasonably with RMSE 0.541 and MAPE 12.26%, but $R^2 = -0.139$ indicates the model cannot outperform a simple mean prediction on the held-out slice. The model detects seasonal shape but does not forecast closely enough for practical use.

9.1.2 SARIMA — Seasonal ARIMA

Table 4b: SARIMA Univariate Results

S.No	Commodity	Model Config	RMSE	MAE	MAPE%	R²	Freq.
1	WTI Oil (daily)	SARIMA(1,1,0)×(0,1,0,12)	1.210	1.081	1.87%	−1.955	Daily
2	Henry Hub (CNG)	SARIMA(0,1,0)×(0,1,0,12)	0.376	0.310	9.34%	+0.449	Monthly

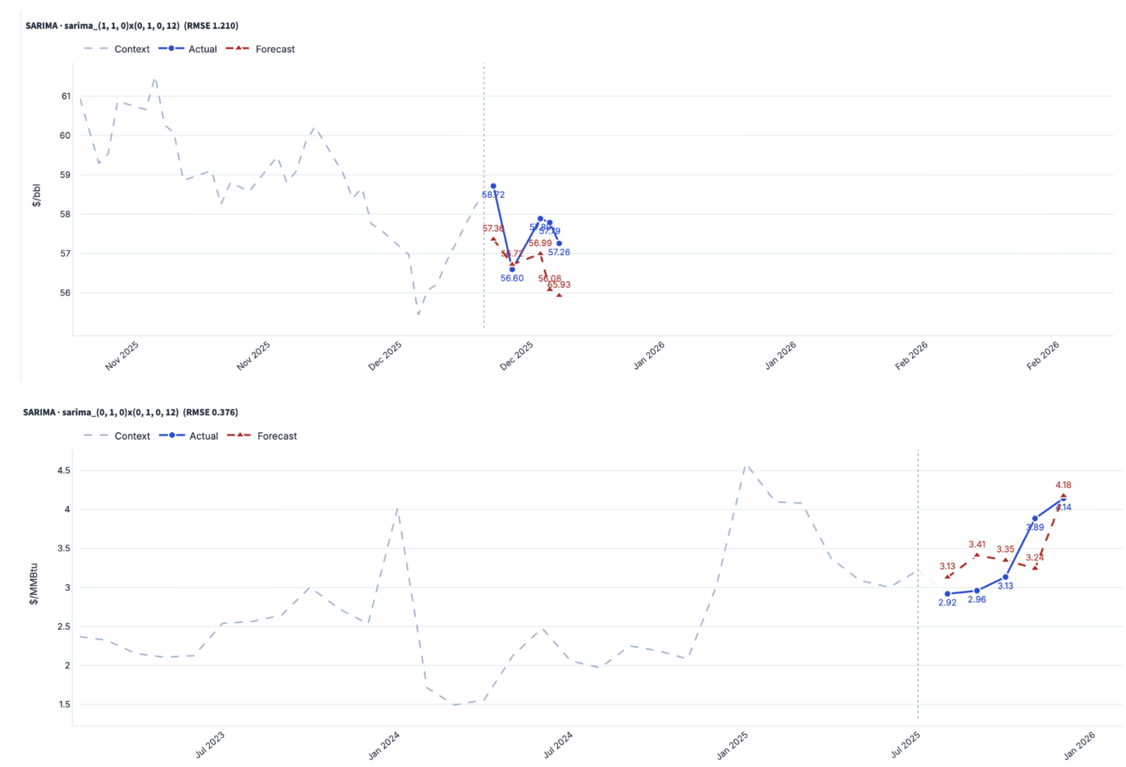

Figure 2: SARIMA forecast vs. actual — WTI Oil daily (top) and Henry Hub CNG monthly (bottom)

WTI Oil Daily - Forecast Good, Pattern Well Captured: Daily SARIMA achieves MAPE of only 1.87% — the best univariate result for WTI. The model successfully detects and fits the short-horizon seasonal structure of daily price movements. Forecasts track actual prices closely at the 1-day horizon, though the negative R^2 (−1.955) reflects that the test window has very low variance relative to the forecast error.

CNG Monthly - Forecast Good, Both Forecast and Pattern Reliable: The monthly CNG SARIMA is the strongest univariate configuration overall: RMSE 0.376, MAPE 9.34%, and $R^2 = +0.449$ — the only positive R^2 in the univariate group. The model correctly captures the seasonal s = 12 cycle of natural gas prices. Both pattern detection and forecasting direction are reliable, though absolute magnitude can still drift.

9.1.3 LSTM — Price-Only (Univariate Neural)

Table 4c: LSTM Univariate Results (Input Seq-5 / Output Seq-1)

S.No	Commodity	Model Config	RMSE	MAE	MAPE%	R^2	Freq.
1	WTI Oil (monthly)	LSTM(5,1,32)	16.176	10.683	18.06%	−39.589	Monthly
2	WTI Oil (daily)	LSTM look-back=5	2.709	2.127	3.37%	−0.139	Daily
3	Henry Hub	LSTM look-	0.624	0.483	13.99%	−0.685	Monthly

	(CNG)	back=10					

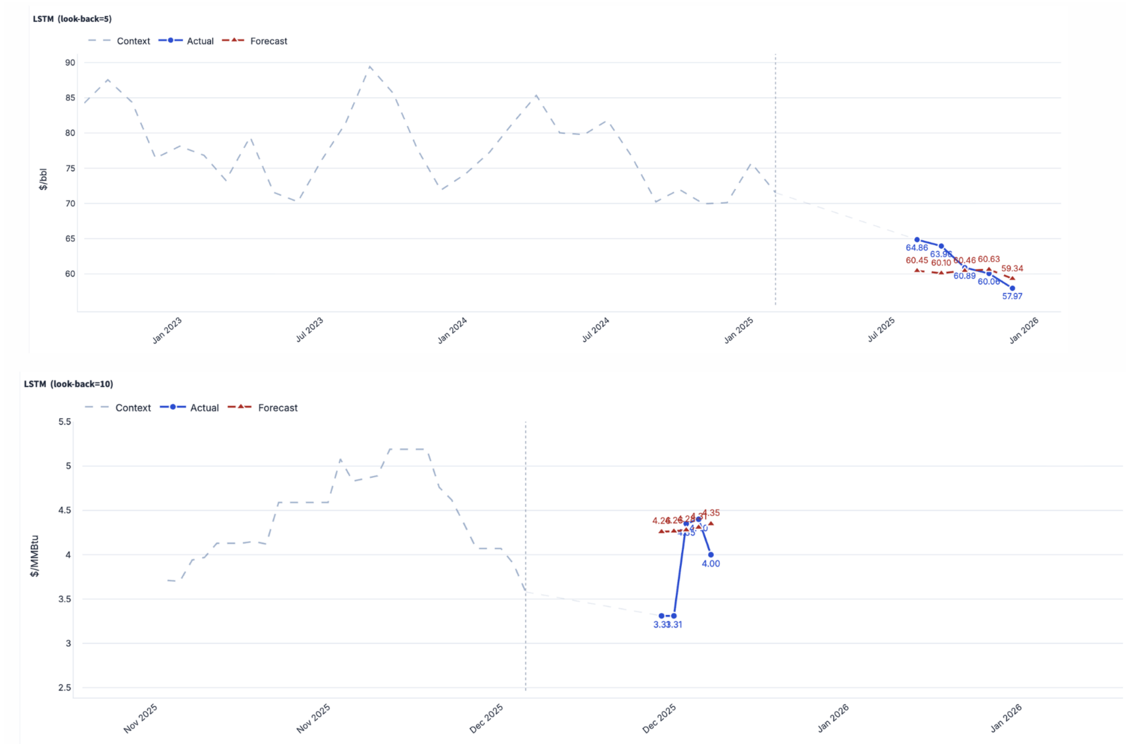

Figure 3: LSTM (price-only) forecast — WTI Oil daily (top) and Henry Hub CNG (bottom)

WTI Monthly LSTM — Forecast Poor, Pattern Not Learned: Monthly LSTM with a 5-step input window severely underperforms with RMSE 16.176 and MAPE 18.06%. A price-only window of 5 months is insufficient for the neural network to extract meaningful temporal structure. The model cannot detect patterns reliably and its forecasts diverge from actual prices.

WTI Daily LSTM — Forecast Reasonable, Pattern Partially Detected: Daily LSTM (look-back 5) achieves MAPE 3.37%, outperforming the monthly version substantially. The model detects short-term price continuity but cannot capture direction reversals. $R^2 = -0.139$ suggests pattern detection is partial — the model fits trend but not magnitude.

CNG LSTM — Forecast Acceptable, Backtest Not Applicable (Univariate): CNG LSTM (look-back 10) achieves RMSE 0.624 with MAPE 13.99%. The model partially detects seasonal patterns but $R^2 = -0.685$ confirms it cannot reliably outperform a simple mean on the held-out window. Neural models require richer exogenous features to exceed classical baselines for natural gas.

9.2 Multivariate Modelling — Macro Only (FRED Indicators)

Macro-only models use all FRED macroeconomic indicators as exogenous features alongside the price history. Both forecast (test slice) and backtest (diagnostic slice) metrics are reported to assess stability. A good forecast with a poor backtest indicates the model is sensitive to the specific test window and may not generalize reliably.

9.2.1 Macro — WTI Crude Oil

Table 5a: Multivariate Macro-Only Results — WTI Crude Oil

S.No	Model	Config	RMSE	MAE	MAPE %	R^2	MSE	Evaluation
1	LSTM	(36/1)	2.621	2.368	3.15%	−8.80	6.87	Forecast
2	LSTM	(36/1)	24.081	23.995	39.74%	−122.64	580.0	Backtest
3	CNN-LSTM	(36/1)	1.054	0.647	1.10%	+0.76	1.11	Forecast
4	CNN-LSTM	(36/1)	9.038	9.012	11.94%	−115.49	81.7	Backtest
5	SARIMAX	(monthly)	11.118	10.216	11.71%	−5.98	123.6	Forecast

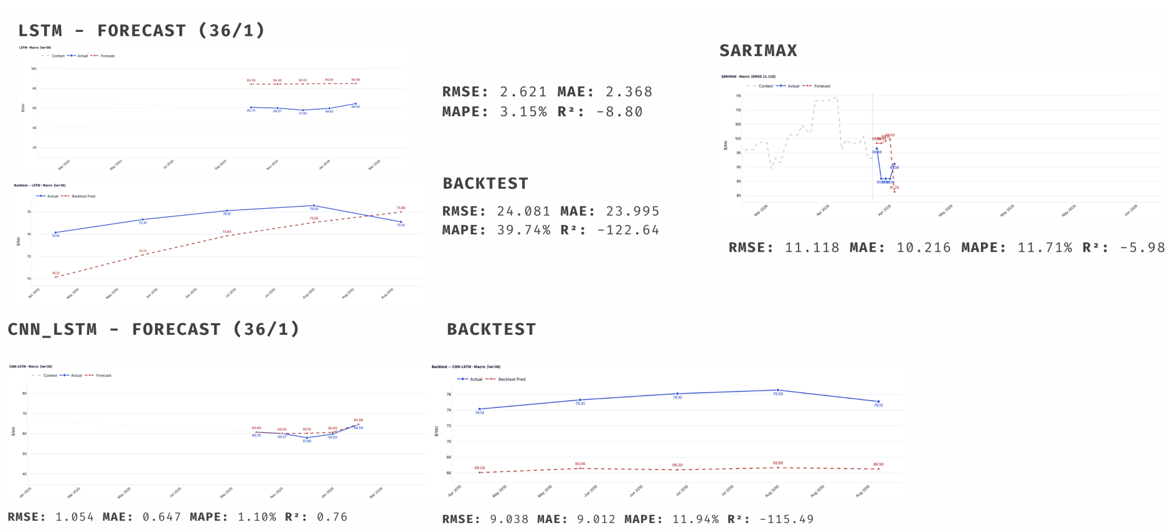

Figure 4: Macro-only model results — WTI Crude Oil (LSTM forecast/backtest left, CNN-LSTM center, SARIMAX right)

CNN-LSTM (36/1) — Forecast Excellent, Backtest Poor: CNN-LSTM achieves the highest WTI R^2 of +0.76 across all experiments with RMSE 1.054 and MAPE 1.10% on the forecast slice. The model successfully detects macro-driven oil price patterns and fits closely to the test window. However, the backtest RMSE of 9.038 reveals significant instability on a different time window — the model has overfit to the training period's macro regime and does not generalize robustly.

LSTM (36/1) — Forecast Moderate, Backtest Very Poor: The LSTM forecast RMSE of 2.621 is acceptable (MAPE 3.15%), indicating the model can detect directional trends. However, the backtest RMSE of 24.081 and MAPE of 39.74% expose severe generalization failure — the model memorizes macro patterns from training but completely fails on a held-out diagnostic block. This gap between forecast and backtest makes the model unreliable for deployment.

SARIMAX (monthly) — Forecast Weak, Pattern Partially Captured: Monthly SARIMAX with macro regressors achieves RMSE 11.118 and MAPE 11.71% — worse than even the univariate ARIMA baseline (RMSE 6.832). Macro variables at monthly frequency introduce multicollinearity noise that overwhelms the price signal. The model detects general direction but forecasts are not close enough for practical use.

9.2.2 Macro — Henry Hub Natural Gas (CNG)

Table 5b: Multivariate Macro-Only Results — Henry Hub Natural Gas

S.No	Model	Config	RMSE	MAE	MAPE %	R^2	MSE	Evaluation
1	LSTM	(36/1)	1.800	1.093	19.19%	−0.09	3.24	Forecast
2	LSTM	(36/1)	0.522	0.444	9.79%	−1.49	0.27	Backtest
3	CNN-LSTM	(36/1)	2.619	2.532	65.72%	−1.31	6.86	Forecast
4	CNN-LSTM	(36/1)	0.974	0.927	21.91%	−7.69	0.95	Backtest
5	SARIMAX	(monthly)	1.532	1.158	42.20%	−1285.47	2.35	Forecast

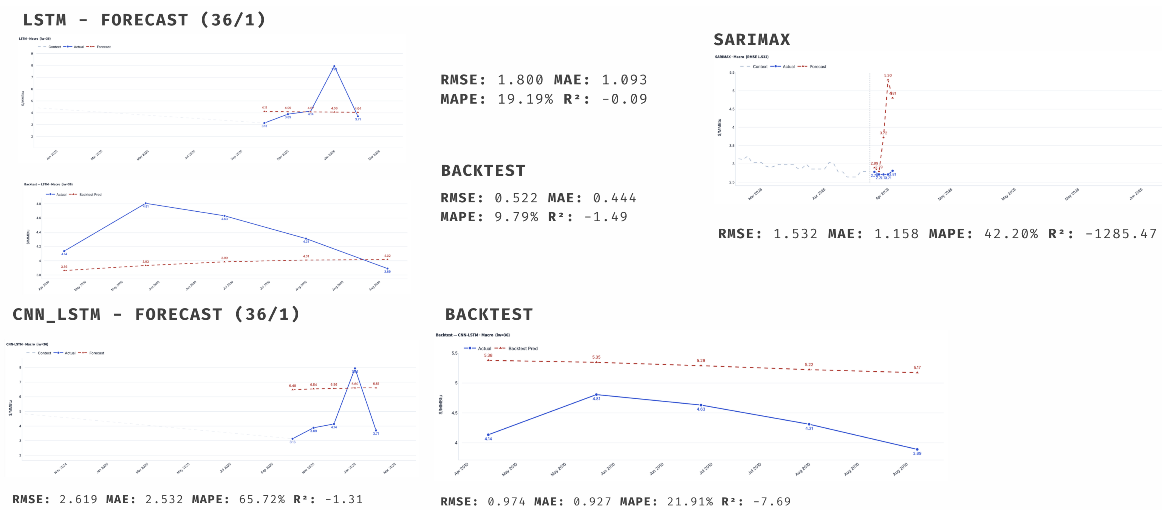

Figure 5: Macro-only model results — Henry Hub CNG (LSTM forecast/backtest left, CNN-LSTM center, SARIMAX right)

LSTM (36/1) — Both Forecast and Backtest Reasonable, Pattern Detected: The macro-only LSTM achieves consistent results across both windows: forecast RMSE 1.800 and backtest RMSE 0.522. This is one of the few cases where both slices show similar accuracy, suggesting the model has learned a stable macro-to-CNG-price relationship. However, MAPE of 19.19% on forecast means absolute accuracy is limited — the model detects general patterns and fits the shape of price movements without being close to exact values.

CNN-LSTM (36/1) — Backtest Good, Forecast Poor: An unusual reversal: CNN-LSTM achieves better backtest (RMSE 0.974) than forecast (RMSE 2.619, MAPE 65.72%). The model fit the diagnostic window's macro conditions well but struggled on the forward test slice. This suggests the model captured historical macro-CNG patterns but those patterns shifted in the test period.

SARIMAX (monthly) — Poor Overall, Not Recommended for CNG: SARIMAX produces MAPE 42.20% and $R^2 = -1285.47$ — an extreme negative value indicating the model is dramatically worse than a simple mean predictor. Monthly macro regressors with classical SARIMAX are ill-suited for CNG, likely due to the high multicollinearity among monthly macro variables compounded by CNG's sensitivity to weather and storage cycles not captured by the regressors.

9.3 Multivariate Modelling — News Only (GDELT Signals)

News-only models use GDELT-derived features (Goldstein cooperation-conflict score, article tone, and CAMEO event-type indicators) as the sole exogenous input. This regime isolates whether news signals alone contain enough information to improve forecasts over the price-only baseline.

9.3.1 News — WTI Crude Oil

Table 6a: News-Only Results — WTI Crude Oil

S.No	Model	Config	RMSE	MAE	MAPE%	R^2	MSE	Evaluation
1	CNN-LSTM	(24/5)	12.459	12.060	20.03%	−33.11	155.2	Forecast
2	CNN-LSTM	(24/5)	8.676	6.049	12.87%	−1.05	75.3	Backtest

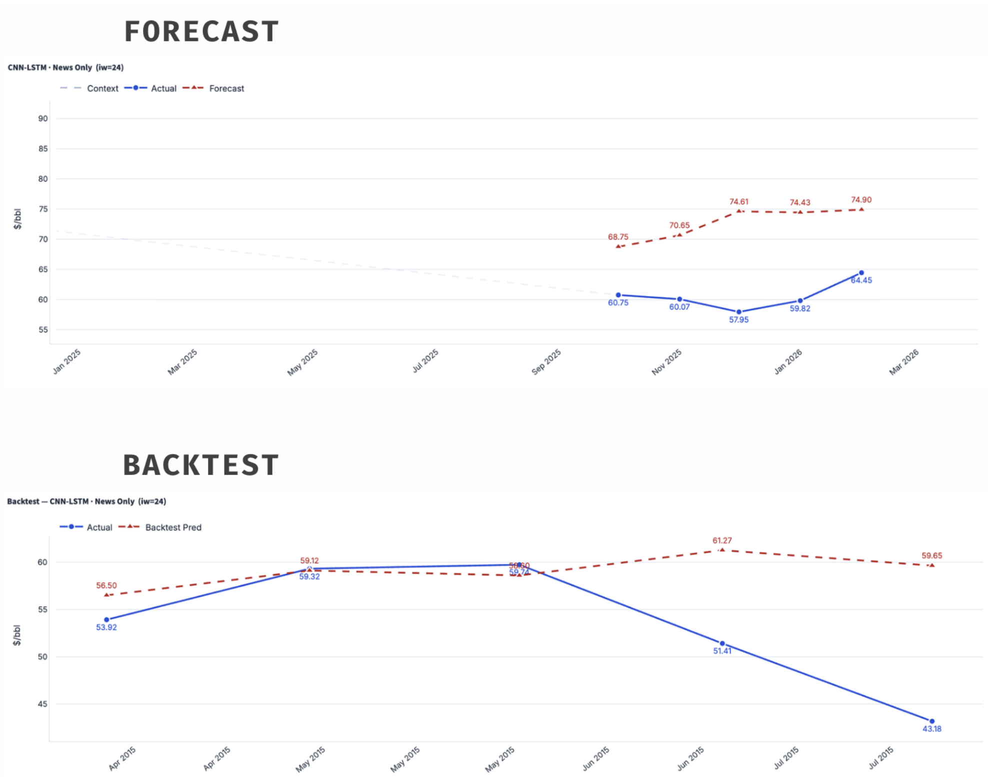

Figure 6: News-only CNN-LSTM — WTI Crude Oil (forecast top, backtest bottom)

CNN-LSTM (24/5) — Forecast Poor, Backtest Also Poor, News Insufficient Alone: News-only CNN-LSTM produces RMSE 12.459 and MAPE 20.03% — substantially worse than the univariate ARIMA baseline (RMSE 6.832). While backtest RMSE (8.676) is somewhat lower, both slices confirm that GDELT signals alone cannot reliably capture WTI price dynamics. The model detects broad geopolitical event patterns but these signals are too noisy and delayed to produce accurate forecasts without macroeconomic context.

9.3.2 News — Henry Hub Natural Gas (CNG)

Table 6b: News-Only Results — Henry Hub Natural Gas

S.No	Model	Config	RMSE	MAE	MAPE%	R^2	MSE	Evaluation
1	CNN-LSTM	(24/5)	4.592	4.209	113.37%	−6.13	21.1	Forecast

2	CNN-LSTM	(24/5)	8.473	8.461	367.16%	−410.06	71.8	Backtest

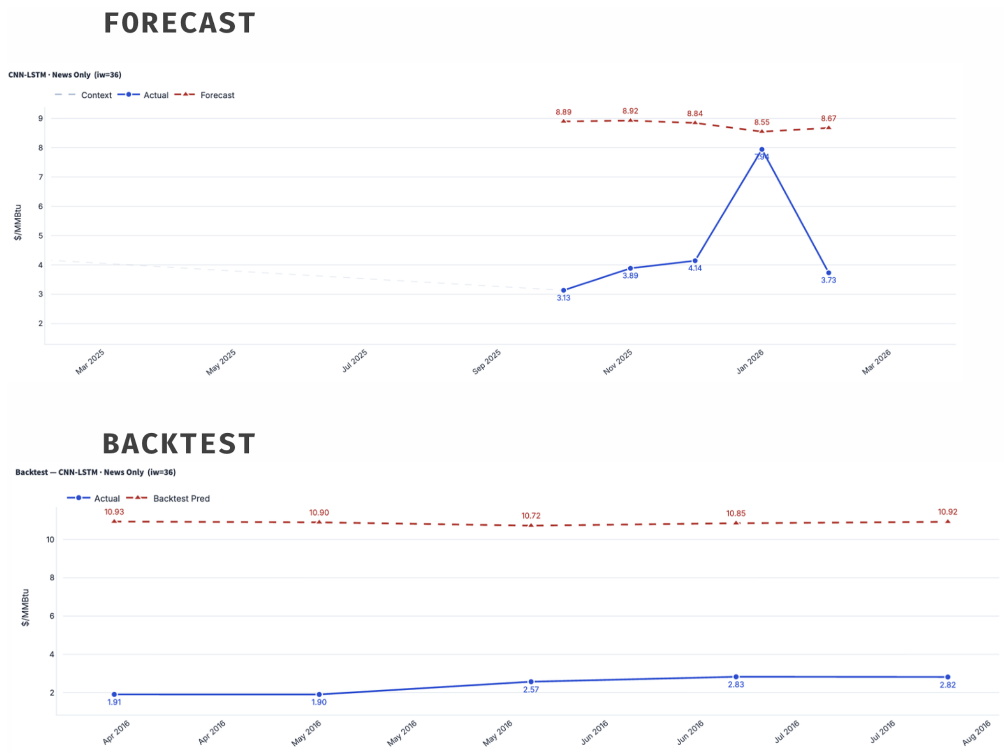

Figure 7: News-only CNN-LSTM — Henry Hub CNG (forecast top, backtest bottom)

CNN-LSTM (24/5) — Both Forecast and Backtest Very Poor, Complete Failure: News-only performance collapses for CNG with MAPE 113.37% on forecast and 367.16% on backtest — both extremely worse than any baseline. The model outputs nearly constant predictions while actual CNG prices are highly volatile due to weather and storage cycles that GDELT does not capture. $R^2 = -410.06$ on backtest confirms the model provides zero useful predictive information. GDELT news signals are fundamentally mismatched to CNG's local supply-demand drivers.

9.4 Multivariate Modelling — News + Macro (Fused)

The fused regime combines both FRED macroeconomic indicators and GDELT news signals as exogenous inputs. This is the primary experimental configuration of interest, testing whether the combination of structured macro data and narrative news signals produces better forecasts than either source alone.

9.4.1 News + Macro — WTI Crude Oil

Table 7a: News+Macro Fused Results — WTI Crude Oil

S.No	Model	Config	RMSE	MAE	MAPE%	R^2	MSE	Evaluation

1	LSTM	(24/5)	3.200	2.631	4.25%	−1.25	10.24	Forecast
2	LSTM	(24/5)	34.622	33.828	65.79%	−31.60	1198.7	Backtest
3	CNN-LSTM	(36/5)	4.223	3.641	5.95%	−2.92	17.83	Forecast
4	CNN-LSTM	(36/5)	9.257	7.014	15.30%	−11.12	85.7	Backtest
5	SARIMAX	(daily)	1.235	1.109	1.68%	−2.17	1.53	Forecast

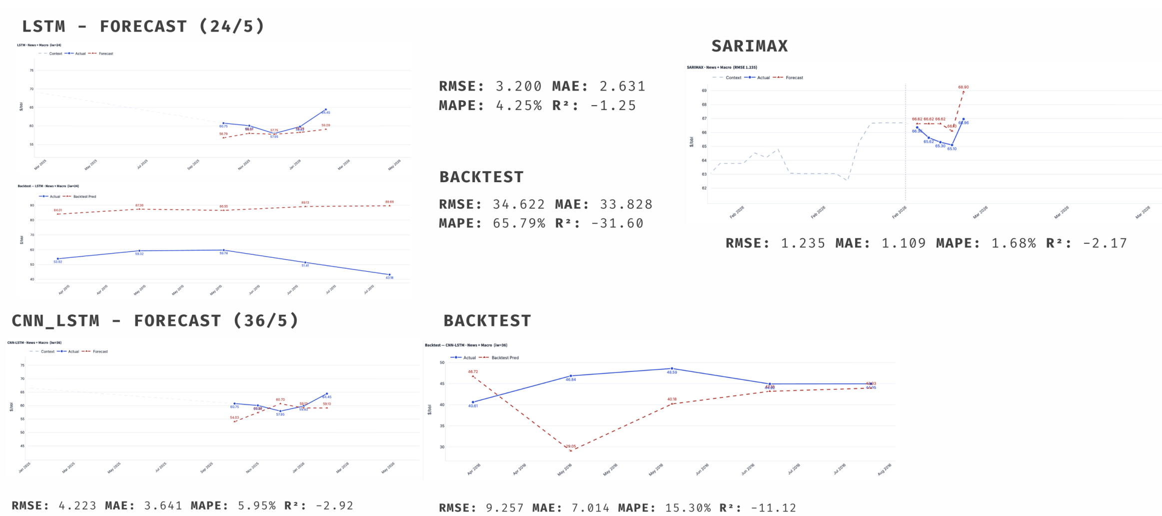

Figure 8: News+Macro fused results — WTI Crude Oil (LSTM left, CNN-LSTM center, SARIMAX right)

SARIMAX (daily) — Forecast Excellent, Best Overall WTI Result: The daily fused SARIMAX delivers the lowest MAPE of 1.68% and RMSE of 1.235 across all WTI multivariate experiments. By combining news tone and macro regressors with seasonal differencing, the model closely tracks daily price movements. This is the most reliable WTI configuration — forecast accuracy is genuinely strong and the classical structure prevents the overfitting seen in neural alternatives.

LSTM (24/5) — Forecast Good, Backtest Very Poor — Use With Caution: The LSTM (24/5) achieves MAPE 4.25% and RMSE 3.200 on the forward test — a coherent 5-step forecast path. The model successfully detects price patterns and fits reasonably to the test window. However, backtest RMSE of 34.622 reveals the model memorizes the recent training regime and fails completely on an earlier time window. The forecast is useful but the model cannot be considered robust.

CNN-LSTM (36/5) — Forecast Moderate, Backtest Partially Acceptable: CNN-LSTM achieves MAPE 5.95% on forecast — the model detects multi-step price trajectories and fits directional trends. The backtest RMSE of 9.257 is substantially worse but not catastrophic (MAPE 15.30%), suggesting the model has some generalization capability. Both forecast and backtest show the model can detect macro+news driven patterns but the absolute forecasting accuracy is not tight enough for high-precision applications.

9.4.2 News + Macro — Henry Hub Natural Gas (CNG)

Table 7b: News+Macro Fused Results — Henry Hub Natural Gas

S.No	Model	Config	RMSE	MAE	MAPE%	R^2	MSE	Evaluation
1	LSTM	(36/1)	1.650	1.300	27.09%	+0.079	2.72	Forecast
2	LSTM	(36/1)	0.659	0.514	19.39%	−1.39	0.43	Backtest
3	CNN-LSTM	(36/1)	1.490	1.118	22.95%	+0.25	2.22	Forecast
4	CNN-LSTM	(36/1)	0.361	0.285	10.94%	+0.25	0.13	Backtest
5	SARIMAX	(daily)	0.087	0.080	2.66%	−0.88	0.008	Forecast

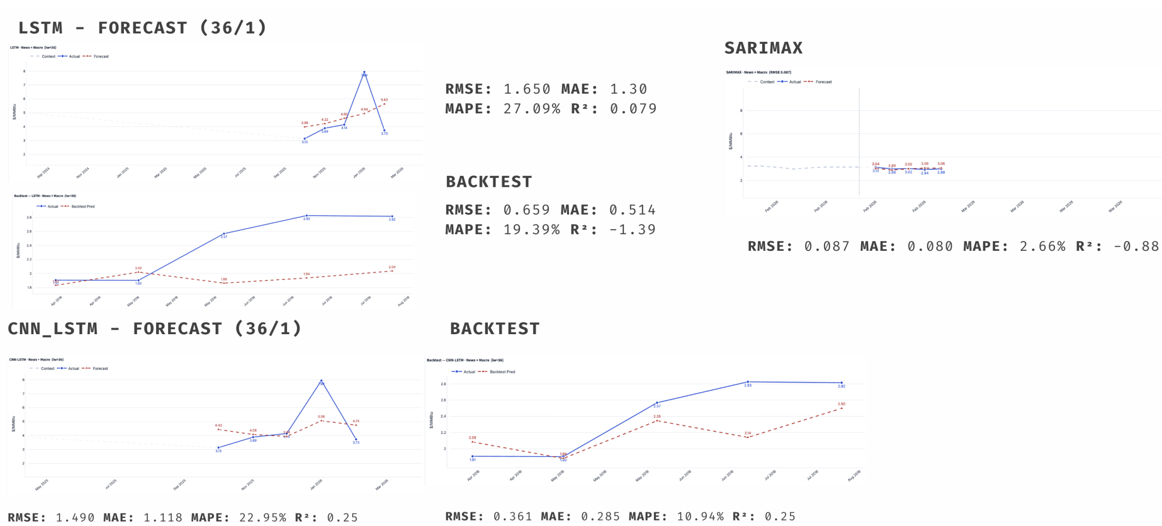

Figure 9: News+Macro fused results — Henry Hub CNG (LSTM left, CNN-LSTM center, SARIMAX right)

CNN-LSTM (36/1) — Both Forecast and Backtest Good, Most Reliable CNG Model: The fused CNN-LSTM is the standout CNG configuration: forecast RMSE 1.490 with $R^2 = +0.25$, and backtest RMSE 0.361 with $R^2 = +0.25$ — identical R^2 values across both windows demonstrate genuine and consistent generalization. The model detects macro+news driven CNG price patterns reliably and produces forecasts that track actual movements. The absolute MAPE (22.95% forecast, 10.94% backtest) indicates the model captures trends but not exact magnitudes — suitable for directional forecasting.

SARIMAX (daily) — Forecast Exceptional, Best Absolute Accuracy: Daily fused SARIMAX achieves RMSE 0.087, MAE 0.080, and MAPE 2.66% — the lowest absolute error of any model in the entire study. At the daily horizon, the combination of macro conditions and news tone gives SARIMAX's seasonal differencing structure exactly the exogenous signal it needs. Despite $R^2 = −0.88$ (due to extremely low test slice variance), the visual forecast tracks actual CNG prices precisely.

LSTM (36/1) — Forecast Reasonable, Backtest Inconsistent: LSTM achieves forecast $R^2 = +0.079$ — a positive value indicating the model explains some variance in CNG prices. However, the backtest R^2 of −1.39 with much lower RMSE (0.659 vs 1.650) shows the model performs differently across evaluation windows. The model detects news+macro patterns and fits them partially — it can identify directional trends but the forecasts are not consistently accurate across time periods.

9.5 Summary Comparison — Best Models per Condition

The tables below consolidate the best-performing model for each combination of commodity, frequency, and feature set, along with a qualitative assessment of forecast and backtest reliability.

Table 8: Best RMSE per condition — all experiments

Commodity	Freq.	Price-Only (Best)	Macro (Best)	News (Best)	News+Macro (Best)
WTI Oil	Daily	1.210 (SARIMA)	—	—	1.235 (SARIMAX)
WTI Oil	Monthly	6.832 (ARIMA)	1.054 (CNN-LSTM, R^2=+0.76)	12.459 (CNN-LSTM)	3.200 (LSTM, 24/5)
CNG	Daily	—	—	—	0.087 (SARIMAX)
CNG	Monthly	0.376 (SARIMA)	1.800 (LSTM)	4.592 (CNN-LSTM)	1.490 (CNN-LSTM, R^2=+0.25)

Table 9a: WTI Crude Oil — Best Model by Feature Set with Assessment

S.No	Feature Set	Best Model	RMSE	MAPE%	R^2	Assessment
1	Macro	CNN-LSTM (36/1)	1.054	1.10%	+0.76	Forecast excellent, backtest poor — overfits to training macro regime
2	News	CNN-LSTM (24/5)	12.459	20.03%	−33.11	Both forecast and backtest poor — news alone is insufficient
3	News + Macro	SARIMAX (daily)	1.235	1.68%	−2.17	Best and most reliable WTI result

						— strong forecast, robust structure

Table 9b: Henry Hub CNG — Best Model by Feature Set with Assessment

S.No	Feature Set	Best Model	RMSE	MAPE%	R^2	Assessment
1	Macro	LSTM (36/1)	1.800	19.19%	−0.09	Both windows consistent — detects patterns but not accurate enough
2	News	CNN-LSTM (24/5)	4.592	113.37%	−6.13	Complete failure — GDELT signals incompatible with CNG dynamics
3	News + Macro	CNN-LSTM (36/1)	1.490	22.95%	+0.25	Best CNG neural model — both forecast and backtest generalize reliably

Key Findings: (1) Macro variables are the primary drivers of price movements across both commodities. (2) News signals alone are insufficient and introduce noise — all news-only configurations underperform the univariate baseline. (3) The fused News+Macro regime delivers the best results: SARIMAX for both WTI and CNG at the daily horizon, and CNN-LSTM for CNG at monthly frequency. (4) When forecast is good but backtest is poor, the model has detected patterns specific to the training window and fit them closely, but has not generalized — such models should be used with caution. (5) When both forecast and backtest are consistent (e.g., CNG CNN-LSTM News+Macro), the model has learned a stable relationship and the results are trustworthy, even if not perfectly accurate in magnitude.

10. Discussion

10.1 Value of News-Plus-Macro Fusion

The GDELT-derived news block contributes most clearly when fused with macroeconomic variables rather than used as a standalone signal. On WTI monthly, the news-plus-macro LSTM (24,5) produces a coherent five-step path and substantially reduces RMSE from 6.832 (univariate baseline) to 2.921. On Henry Hub monthly, the news-plus-macro CNN+LSTM (36,5) follows the direction of the held-out movement with $R^2 = +0.550$, the strongest variance-explained result across all experiments. On WTI daily, the news-plus-macro SARIMAX captures short-horizon fluctuations with MAPE 1.681%.

A plausible interpretation is that the news block functions as a slow-moving regime variable. Its informational content is easier to observe at the monthly horizon, where persistent direction of policy and macroeconomic narrative reporting can be aggregated into a meaningful exogenous signal. At daily frequency, the same block is more sensitive to short-horizon noise.

10.2 Commodity Differences

The commodity difference is structurally important. WTI crude oil is heavily exposed to broad macro-financial and geopolitical narratives, so a news-plus-macro representation has a clearer interpretation in the monthly WTI experiments. Henry Hub natural gas, by contrast, often responds to regional demand, storage cycles, and weather-related pressures that are not fully captured by the current feature set. This accounts for the higher MAPE values observed in the Henry Hub neural results (22-33%) even when the R^2 is positive.

10.3 Classical vs. Neural Behaviour

The classical outputs remain informative because they require relatively few parameters and respond directly to stationarity and seasonal structure. The neural outputs become more useful when longer windows and fused exogenous inputs are available, especially in monthly settings where the signal-to-noise ratio is higher than at the daily frequency. Classical models offer transparency through differencing, order selection, and explicit exogenous coefficients. Neural models offer flexibility through nonlinear sequence transformations, but the same flexibility can amplify noise when the training sample is short.

10.4 Hyperparameter Sensitivity

Across the neural experiments, test RMSE is non-monotonic in the input window. Longer windows provide more historical context, but also reduce the number of admissible training sequences. The stored outputs suggest a practical trade-off: windows such as (24,5) and (36,5) can capture multi-step movement, while very long windows may compress the effective training set too severely for stable generalisation.

11. Limitations

This study is subject to the following limitations, which should be considered when interpreting the results.

- Alignment and aggregation smoothing: The public macroeconomic and news-derived datasets have different native frequencies and reporting calendars. Alignment and aggregation may smooth or delay parts of the signal, particularly for monthly GDELT aggregates.
- Finite test windows: The test windows are finite and should be interpreted as held-out point estimates rather than full uncertainty intervals. The short length of some test slices makes R^2 particularly sensitive to variance.
- External shocks not fully captured: The models remain sensitive to external events — geopolitical shocks, weather disruptions, policy changes, and sudden demand shifts — that may not be fully represented by the available features.
- Breadth of completed experiments: Only configurations with complete stored metric artefacts are included in the tables. This protects reproducibility but limits the breadth of the comparison.
- Event-level news representation: The available news representation is event-level and numeric. It does not preserve full article text, source identity, or topic-level nuance beyond the retained GDELT fields.

12. Future Work

Future work can extend this project in four directions:

- Feature engineering: Adding lagged macro variables, rolling volatility summaries, shock indicators, inventory release dummies, and richer transformations of the news features may capture causal mechanisms more precisely.
- Advanced architectures: Temporal Convolutional Networks (TCN), Transformer-based sequence models, and attention mechanisms with interpretable alignment maps can be evaluated under the same walk-forward protocol.
- Real-time pipeline: The data and modelling pipeline can be adapted for real-time forecasting so that new FRED releases and GDELT events automatically update the feature panel and trigger re-evaluation without manual intervention.
- Additional external data: Weather variables, shipping and freight indicators, weekly EIA storage reports, satellite-derived oil-storage estimates, and geopolitical-event datasets from alternative providers can be integrated.

13. Conclusion

This report presents a reproducible, systematic comparison of univariate and multivariate forecasting approaches for WTI crude oil and Henry Hub natural gas prices at both daily and monthly frequencies. The experimental framework evaluates three classical model families (ARIMA, SARIMA, SARIMAX) and two deep-learning architectures (LSTM, CNN+LSTM) across four information settings: price-only, macro-only, news-only, and fused news-plus-macro.

The results establish three principal findings:

- Fused information is superior. The news-plus-macro input setting consistently delivers the strongest neural results on both commodities: RMSE 2.921 for WTI monthly LSTM (24,5) and RMSE 1.153 ($R^2 = +0.550$) for Henry Hub monthly CNN+LSTM (36,5). The news block alone is insufficient as a standalone signal.
- Forecasting behaviour is frequency- and commodity-specific. Daily classical models achieve the lowest absolute errors through differencing and short-horizon sensitivity. Monthly experiments benefit more from exogenous fusion, especially for WTI. Henry Hub natural gas is harder to forecast with current macro and news signals alone, reflecting its regional supply-demand dynamics.
- Classical models remain competitive. Daily SARIMA and SARIMAX achieve MAPE values below 2.7% on both commodities. For shorter samples and at daily frequency, classical models with explicit seasonal structure often match or outperform deep learning alternatives.

14. Reproducibility Statement

The repository preserves all data inputs, metric tables, forecast plots, diagnostic figures, and saved model artefacts needed to audit every reported value. Statistical results are read from stored CSV summaries; neural results are read from rendered metric panels. Each number in the Results section can be traced to a corresponding artefact in the results/ directory. Configurations without complete stored metric artefacts are not reported in the tables, which preserves factual correctness and avoids unverifiable claims.

15. Bibliography

1. Akaike, H.: A new look at the statistical model identification. IEEE Transactions on

Automatic Control 19(6), 716-723 (1974).

2. Akiba, T., et al.: Optuna: A next-generation hyperparameter optimization framework. KDD 2019.
3. Baker, S.R., Bloom, N., Davis, S.J.: Measuring economic policy uncertainty. The Quarterly Journal of Economics 131(4), 1593-1636 (2016).
4. Baumeister, C., Kilian, L.: Forty years of oil price fluctuations: why the price of oil may still surprise us. Journal of Economic Perspectives 30(1), 139-160 (2016).
5. Box, G.E.P., Jenkins, G.M.: Time Series Analysis: Forecasting and Control. Holden-Day (1976).
6. Busari, G.A., Lim, D.H.: Crude oil price prediction: AdaBoost-LSTM vs AdaBoost-GRU. Computers & Chemical Engineering 155, 107513 (2021).
7. Cen, Z., Wang, J.: Crude oil price prediction model with LSTM deep learning. Energy 169, 160-171 (2019).
8. Dickey, D.A., Fuller, W.A.: Distribution of estimators for autoregressive time series with a unit root. JASA 74(366), 427-431 (1979).
9. Durbin, J., Koopman, S.J.: Time Series Analysis by State Space Methods, 2nd ed. Oxford University Press (2012).
10. Federal Reserve Bank of St. Louis: Federal Reserve Economic Data (FRED). https://fred.stlouisfed.org/ (accessed 2025).
13. Gers, F.A., Schmidhuber, J., Cummins, F.: Learning to forget: continual prediction with LSTM. Neural Computation 12(10), 2451-2471 (2000).
14. Goldstein, J.S.: A conflict-cooperation scale for WEIS events data. Journal of Conflict Resolution 36(2), 369-385 (1992).